

1. Set up the partial fraction decomposition of the following rational functions. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - 2x^2 + 3x - 4}{x(x-1)^2(x^2+9)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2} + \frac{Dx+E}{x^2+9} + \frac{Fx+G}{(x^2+9)^2}$$

2. Evaluate the improper integral $\int_1^{\infty} \frac{\arctan x}{1+x^2} dx$ or show that it is divergent.

(Note: Answers without appropriate limit notations will result in loss of points.)

$$u = \arctan x, \quad du = \frac{1}{1+x^2} dx$$

$$\int \frac{\arctan x}{1+x^2} dx = \int u du = \frac{1}{2} u^2 + C = \frac{1}{2} (\arctan x)^2 + C$$

$$\int_1^{\infty} \frac{\arctan x}{1+x^2} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{\arctan x}{1+x^2} dx = \lim_{t \rightarrow \infty} \left(\frac{1}{2} (\arctan x)^2 \Big|_1^t \right)$$

$$= \lim_{t \rightarrow \infty} \left[\frac{1}{2} (\arctan t)^2 - \frac{1}{2} (\arctan 1)^2 \right]$$

$$= \lim_{t \rightarrow \infty} \left[\frac{1}{2} \left(\frac{\pi}{2} \right)^2 - \frac{1}{2} \left(\frac{\pi}{4} \right)^2 \right]$$

$$= \frac{1}{2} \cdot \frac{\pi^2}{4} - \frac{1}{2} \cdot \frac{\pi^2}{16}$$

$$= \frac{1}{8} \pi^2 - \frac{1}{32} \pi^2 = \frac{3}{32} \pi^2$$

Conclusion: $\int_1^{\infty} \frac{\arctan x}{1+x^2} dx = \frac{3}{32} \pi^2$, Converges.